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Total Syntheses of (−)-Conidiogenone B, (−)-Conidiogenone, and (−)-Conidiogenol

*Bo Xu[+], Wen Xun[+], Shaobin Su, and Hongbin Zhai**

In memory of Professor Jack E. Baldwin

Abstract: *Cyclopianes are novel diterpenes featuring a highly strained 6/5/5/5 tetracyclic core embedded with 6–8 consecutive stereocenters. The concise total syntheses of (−)-conidiogenone B, (−)-conidiogenone, and (−)-conidiogenol have been accomplished in 14–17 steps. The present work features a HAT-mediated alkene–nitrile cyclization to access the cis-biquinane, a Nicholas/Pauson–Khand reaction to construct the linear triquinane, and a Danheiser annulation to afford the congested angular triquinane skeleton.*

Cyclopianes are novel diterpenes featuring a highly strained 6/5/5/5 tetracyclic core embedded with 6–8 consecutive stereocenters, four of which are quaternary carbon centers (Figure 1). The structural variations of cyclopianes are mainly manifested as differences in the oxidation state of certain carbon atoms within the skeleton. Since the first isolation of conidiogenone (**1**) and conidiogenol (**2**) by Sterner and co-workers in 2002, more than 20 compounds of this family have been isolated from the genus *Penicillium*.[1] Interestingly, cyclopiane diterpenes display a range of biological activities. Conidiogenone (**1**) and conidiogenol (**2**) as inducers of conidiogenesis in *P. cyclopium* may become new tool compounds for the study of the morphogenetic event.[1a] Conidiogenone B (**3**) shows significant antibacterial activity against methicillin-resistant *Staphylococcus aureus* (MRSA; MIC = 8 μg mL^{-1}).[1d] Conidiogenone C (**4**) exhibits potent cytotoxicity against HL60 cells (IC_{50} = 38 nM).[1c]

The structural complexity and the significant biological activities of cyclopianes have aroused broad interest from the synthetic community.[2] In 2016, Tu and co-workers accomplished the first total syntheses of cyclopiane diterpenoids, featuring a one-step regioselective and diastereoselective semipinacol-type rearrangement strategy.[2a] In 2019, Snyder and co-workers reported the asymmetric total syntheses of cyclopiane diterpenoids guided by quaternary center analysis.[2b] In addition to the chemical synthesis, the biosynthetic pathway of conidiogenone was fully elucidated through heterologous expression of biosynthetic genes in *Aspergillus oryzae* and by in vitro enzyme assay with ^{13}C-labeled substrates in 2019.[3]

The synthesis of angular triquinane natural products has continued to be an active research area in organic chemistry. In the past decades, many novel synthetic strategies for angular triquinane skeleton natural products have been developed.[4] Inspired by Schmidt and Knölker's work,[5] which utilized a Lewis acid mediated allylsilane [3+2] cycloaddition strategy to forge the angular triquinane skeleton, we became fascinated with the development of a general Danheiser annulation strategy toward the construction of angular triquinane natural products including silphinene (**7**), retigeranic acid A (**8**), and crinipellin A (**9**; Figure 2A). Danheiser annulation of allenylsilanes with electron-deficient olefins is a powerful and straightforward method for the synthesis of highly substituted cyclopentenes.[6] However, this annulation has received little attention from synthetic organic chemists since its first publication by Danheiser in 1981. In particular, it has not been extensively examined for the synthesis of complex natural products.[7] Herein, we report the total syntheses of (−)-conidiogenone B, (−)-conidiogenone, and (−)-conidiogenol featuring a Danheiser annulation reaction for the rapid assembly of the challenging tetracyclic skeleton of these natural products.

The retrosynthetic analysis is outlined in Figure 2B. We envisioned that (−)-conidiogenone (**1**) and (−)-conidiogeno-

[*] Dr. B. Xu,[+] Dr. W. Xun,[+] S. Su, Prof. Dr. H. Zhai
The State Key Laboratory of Chemical Oncogenomics, Guangdong Provincial Key Laboratory of Nano-Micro Materials Research, School of Chemical Biology and Biotechnology, Shenzhen Graduate School of Peking University, Shenzhen 518055 (China)
E-mail: zhaihb@pku.edu.cn
Prof. Dr. H. Zhai
Institute of Marine Biomedicine, Shenzhen Polytechnic
Shenzhen 518055 (China)
and
Collaborative Innovation Center of Chemical Science and Engineering (Tianjin), Tianjin 300071 (China)
[+] These authors contributed equally to this work.
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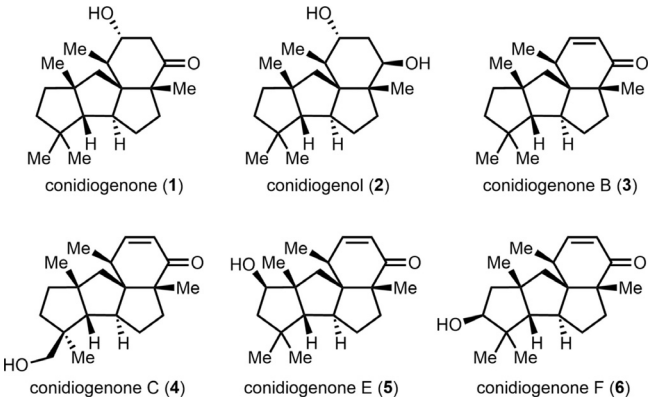


Figure 1. Representative structures of cyclopiane-type diterpenes.

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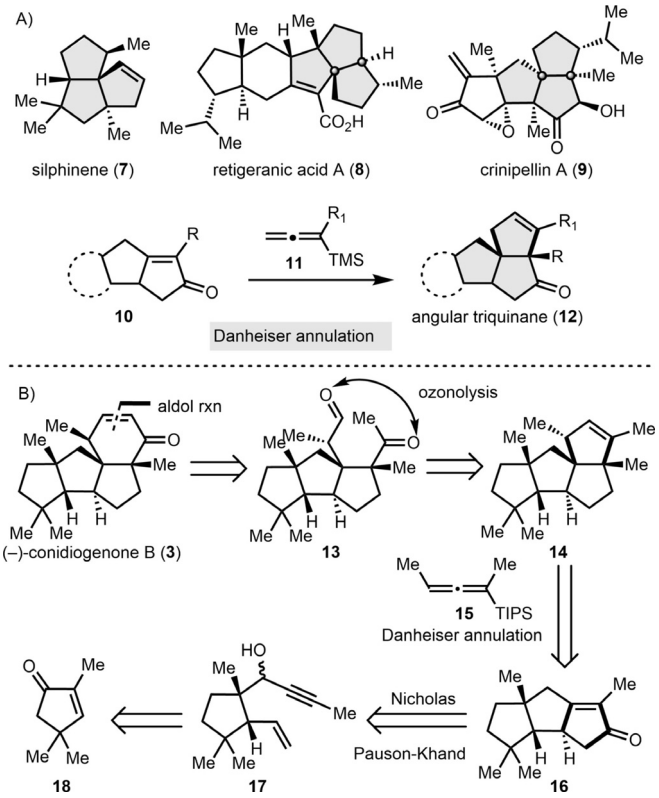


Figure 2. A) Danheiser annulation to access angular triquinane. B) Retrosynthetic analysis of (−)-conidiogenone B (**3**).

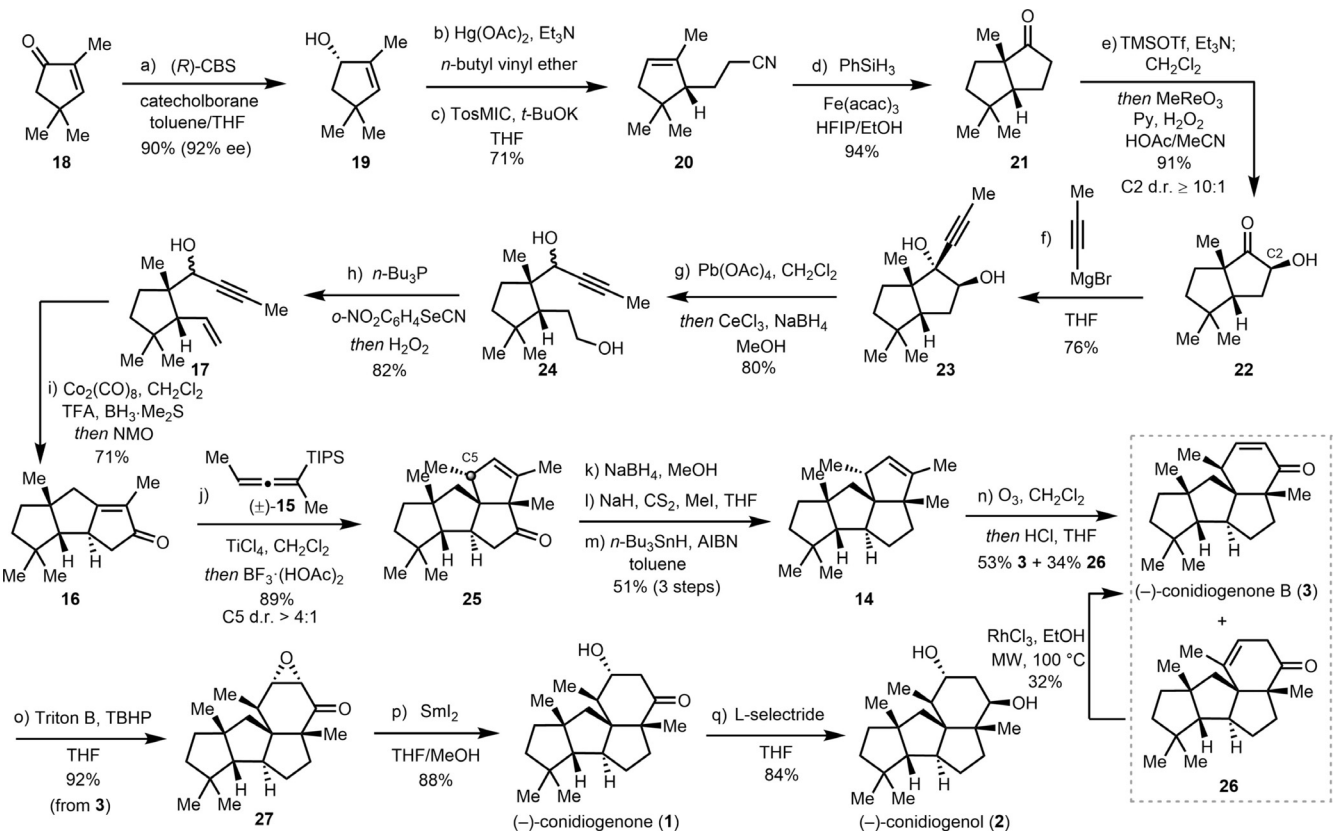
ne B (**3**) could be constructed through an aldol approach from keto aldehyde **13**, which could be obtained by ozonolysis of the trisubstituted olefin **14**. The congested angular triquinane portion of the tetraquinane **14** could be accessed efficiently through a [3+2] Danheiser annulation, and the key linear triquinane **16** could be readily derived via a tandem Nicholas/Pauson–Khand reaction sequence from **17**, which in turn could be traced back to trimethylcyclopentenone **18**.[8]

As outlined in Scheme 1, our synthesis commenced with the Corey–Bakshi–Shibata reduction[9] of the readily available trimethylcyclopentenone **18**,[8] providing the corresponding allylic alcohol **19** in 90% yield with 92% *ee* (as determined by the HPLC analysis of the corresponding *p*-nitrobenzoate). Treatment of allylic alcohol **19** with *n*-butyl vinyl ether in the presence of $Hg(OAc)_2$ and Et_3N at 170 °C completed a tandem vinylation/Claisen rearrangement reaction sequence[10] to afford the desired aldehyde, one-carbon homologation of which through van Leusen reaction[11] furnished nitrile **20** in 71% yield. The iron-mediated HAT alkene–nitrile cyclization of **20** under Talbot's conditions provided bicyclic ketone **21** (74%).[12] Examination of several conventional methods (such as TMSOTf/*m*-CPBA, LDA/Davis oxaziridine, LDA/MoOPH) for achieving the α-hydroxylation ketone **22** proved problematic. Fortunately, $MeReO_3$-catalyzed Rubottom oxidation was effective for this transformation and the reaction led reproducibly to the desired product **22** in high yield (91%, ≥10:1 d.r.).[13] Addition of 1-propynyl Grignard to **22** afforded the vicinal diol **23** in 76% yield, oxidative cleavage of which by using $Pb(OAc)_4$ followed by direct reduction of both carbonyl groups under Luche conditions generated **24** in 80% yield. Chemoselective transformation of the primary hydroxy group in **24** to give **17** (82%) through Grieco's method[14] set the stage for the key Nicholas/Pauson–Khand reaction sequence. Even though the combination of Nicholas and Pauson–Khand reactions into a one-pot fashion would enable synthetic efficiency enhancement, the sole application of this strategy that had been reported was Schreiber's synthesis of epoxydictymene.[15] Gratifyingly, enyne alcohol **17** was subjected to a Nicholas reaction with borane-methyl sulfide complex as the hydride source,[16] followed by amine-N-oxide-promoted Pauson–Khand reaction,[17] to furnish triquinane **16** as a single diastereomer in 71% yield. To the best of our knowledge, this represents the first example of tandem Nicholas/Pauson–Khand reactions involving a hydride reduction step.

We next turned our attention to the key Danheiser annulation to forge a five-membered ring with concomitant establishment of two all-carbon quaternary carbon centers. Treatment of **16** and commercial allene **15a** with $TiCl_4$ at −78 °C afforded the desired tetraquinane **28a** bearing two quaternary carbons in 91% yield. Encouraged by this result, we envisioned that allene **15b** would also be effective for the reaction. Indeed, the cyclization did take place to generate tetraquinane **28b** in 33% yield along with the competing Hosomi–Sakurai product **29b** in 46% yield. The Sakurai-type product presumably resulted from insufficient stability of the intermediary vinyl cation. We speculated that increasing the bulkiness of the silyl group in the allenylsilane might prevent the formation of the Sakurai-type byproduct. Thus, a bulky TIPS group was employed to replace TMS for allene **15c**, and in this case the desired **28c** was obtained in a much better yield (89%), together with less than 5% of the Sakurai-type byproduct **29c**. Subsequent one-pot desilylation of **28c** with a boron trifluoride acetic acid complex provided the desired tetraquinane **25** in 89% yield (d.r. >4:1). The relative stereochemistry of **25** was established through 2D NMR spectroscopy and further corroborated by X-ray analysis of the corresponding tosylhydrazone *ent*-**30**. It is reasonable to assume that the transition state conformation *TS*-B was more favored over *TS*-A as shown in Figure 3. At this point, the carbonyl had to be fully reduced to methylene. Wolff–Kishner–Huang-Minlon reduction[18] of ketone **25** was problematic since the desired full reduction product **14** was obtained in only 37% yield. Modified Wolff–Kishner reduction of tosylhydrazone **30** also proved to be ineffective with a variety of different reducing agents such as $NaBH_3CN$, catecholborane, $LiAlH_4$, or Dibal-H.[19] Nevertheless, a three-step Barton–McCombie deoxygenation approach successfully afforded tetracycle **14** in 51% overall yield.[20]

With the requisite angular triquinane framework in hand, we began to investigate the ring-enlargement process. Ozonolysis of the trisubstituted carbon-carbon double bond in tetraquinane **14** provided keto aldehyde **13**, which was directly used for the subsequent aldol reaction. Prior to the planned aldol condensation for the generation of (−)-conidiogenone B (**3**), we first examined the feasibility of direct conversion of keto-aldehyde **13** into (−)-conidiogenone (**1**) through a stereoselective aldol reaction. However, the

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Scheme 1. Total syntheses of (−)-conidiogenone B, (−)-conidiogenone and (−)-conidiogenol. Reagents and conditions: a) (*R*)-CBS (0.2 equiv), catecholborane (1.8 equiv), toluene, −78 °C, 90%, 92% *ee*; b) $Hg(OAc)_2$ (0.15 equiv), Et_3N (0.15 equiv), *n*-butyl vinyl ether, 170 °C; c) TosMIC (1.7 equiv), *t*-BuOK (2.67 equiv), THF, −60 °C; then MeOH, reflux, 71% for 2 steps; d) $Fe(acac)_3$ (0.5 equiv), $PhSiH_3$ (3.1 equiv), EtOH/HFIP, 50 °C; then 2 M HCl, reflux, 94%; e) Et_3N (2 equiv), TMSOTf (1.3 equiv), CH_2Cl_2, 0 °C; $MeReO_3$ (0.03 equiv), 30% H_2O_2 (4 equiv), pyridine (1 equiv), HOAc/MeCN, 0 °C to rt; PPTS (0.5 equiv), rt, 91%; f) 1-propynylMgBr (3 equiv), THF, 0 °C; g) $Pb(OAc)_4$ (1.5 equiv), CH_2Cl_2, rt; then $CeCl_3 \cdot 7H_2O$ (2 equiv), $NaBH_4$ (4 equiv), MeOH, 0 °C, 80%; h) 2-nitrophenylselenocyanate (2 equiv), *n*-Bu_3P (2 equiv), THF, 0 °C to rt; then $NaHCO_3$ (6 equiv) and 30% H_2O_2 (20 equiv), 0 °C to rt, 82%; i) $Co_2(CO)_8$ (1.1 equiv), CH_2Cl_2, 0 °C to rt; then TFA (13.5 equiv), $BH_3 \cdot Me_2S$ (2 equiv), 0 °C; then $NaHCO_3$ (20 equiv), NMO (10 equiv), rt, 71%; j) (±)-**15** (2.5 equiv), $TiCl_4$ (2 equiv), −78 °C to rt; then $BF_3 \cdot (HOAc)_2$ (10 equiv), rt, 89%; k) $NaBH_4$ (10 equiv), MeOH, rt; l) NaH (3 equiv), imidazole (0.1 equiv), CS_2 (3 equiv), MeI (10 equiv); m) AIBN (0.2 equiv), *n*-Bu_3SnH (2.5 equiv), toluene, reflux, 51% for 3 steps; n) O_3, CH_2Cl_2, −78 °C; Me_2S, −78 °C to rt; then 3 M HCl, THF, reflux, 53%; o) Triton B (2 equiv), TBHP (20 equiv), THF, 92%; p) SmI_2 (10 equiv), THF/MeOH, 88%; q) L-selectride (6 equiv), THF, −78 °C, 84%. AIBN = azobisisobutyronitrile, CBS = 2-methyl-CBS-oxazaborolidine, HFIP = hexafluoro-2-propanol, NMO = N-methylmorpholine N-oxide, THF = tetrahydrofuran, TMSOTf = trimethylsilyl trifluoromethane-sulfonate, TosMIC = *p*-toluenesulfonylmethyl isocyanide, Tritone B = benzyltrimethylammonium hydroxide.

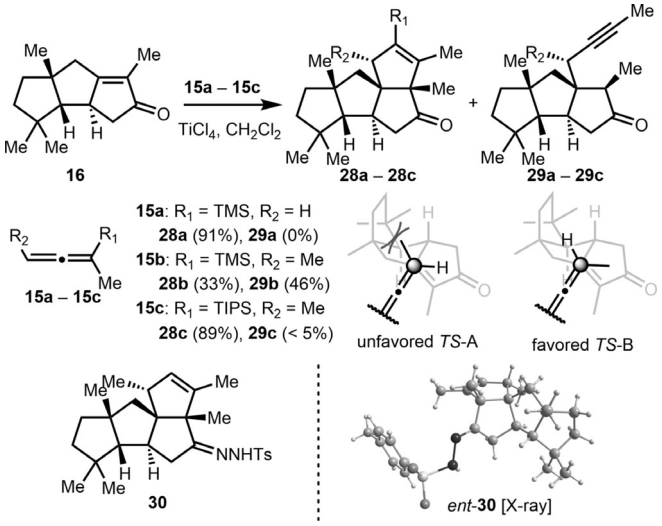
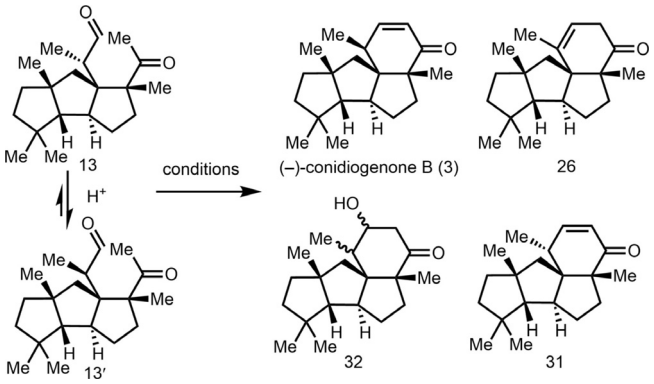


Figure 3. Investigation of the pivotal Danheiser annulation.

reaction failed to lead to (−)-conidiogenone (**1**) despite many attempts (Table 1). Surprisingly, treatment of aldehyde **13** with *p*-toluenesulfonic acid in dry toluene at 80 °C afforded unconjugated enone **26** in 78% yield. After extensive optimization efforts, we were pleased to find that subjecting **13** to 3 M HCl at reflux afforded (−)-conidiogenone B (**3**) in 53% yield together with the isomeric enone **26** in 34% yield. It is worth noting that the concentration of aqueous HCl was critical to avoiding the formation of the undesired product **31**. Isomerization of **26** to (−)-conidiogenone B (**3**) was achieved by treatment with $RhCl_3$ ($K_{eq} \approx 0.5$). Finally, stereoselective epoxidation[2a] of (−)-conidiogenone B with TBHP followed by reduction of the resultant epoxide with SmI_2[21] furnished (−)-conidiogenone (**1**). Reduction of (−)-conidiogenone (**1**) with L-selectride delivered (−)-conidiogenol (**2**) in 84% yield. The spectroscopic data of the synthetic sample match those reported.[1a,c]

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Table 1: Investigation of the aldol condensation conditions.



Entry	Conditions	Yield [%]				
		3	**31**	**26**	**32**[a]	rsm[b]
1	KOH, MeOH, rt	–	–	–	–	–
2	LiOH, *i*PrOH, rt, 48 h	–	46%	–	32%	–
3	DBU, CH_2Cl_2, 24 h	–	–	–	42%	24%
4	L-proline, DMSO, rt to 80 °C	–	–	–	–	35%
5	(±)-BNPPA, CH_2Cl_2, rt	–	–	–	–	32%
6	*p*-TsOH, toluene, 80 °C	–	–	78%	–	–
7	1 M HCl, reflux	37%	31%	–		–
8	2 M HCl, reflux	27%	25%	31%	–	–
9	3 M HCl, reflux	53%	–	34%	–	–

[a] a mixture of four diastereomeric without further separation. [b] **13** and its epimer **13′**. BNPPA = 1,1′-binaphthalene-2,2′-diyl phosphoric acid.

In summary, a concise total synthesis of (−)-conidiogenone B has been achieved in 14 steps and with 4.7% overall yield from readily available trimethylcyclopentenone **18**.[8] Moreover, (−)-conidiogenone and (−)-conidiogenol were also realized through a modified Tu protocol.[2a] The synthesis features a HAT-mediated alkene–nitrile cyclization to access the *cis*-biquinane, a sequential Nicholas/Pauson–Khand reactions for the construction of the linear triquinane, a Danheiser annulation to forge the congested angular triquinane portion, and a combined ozonolysis/aldol reaction to assemble the α,β-unsaturated cyclohexenone skeleton. The current work may facilitate larger-scale preparation and further biological studies of various cyclopiane natural products.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: cyclopianes · Danheiser annulation · diterpenes · Nicholas/Pauson—Khand reaction · total synthesis

[1] a) T. Roncal, S. Cordobés, O. Sterner, U. Ugalde, *Eukaryotic Cell* **2002**, *1*, 823–829; b) T. Roncal, S. Cordobés, U. Ugalde, Y. He, O. Sterner, *Tetrahedron Lett.* **2002**, *43*, 6799–6802; c) L. Du, D. Li, T. Zhu, S. Cai, F. Wang, X. Xiao, Q. Gu, *Tetrahedron* **2009**, *65*, 1033–1039; d) S.-S. Gao, X.-M. Li, Y. Zhang, C.-S. Li, B.-G. Wang, *Chem. Biodiversity* **2011**, *8*, 1748–1753; e) S.-S. Gao, Z. Shang, X.-M. Li, C.-S. Li, C.-M. Cui, B.-G. Wang, *Biosci. Biotechnol. Biochem.* **2012**, *76*, 358–360; f) S. Niu, Z.-W. Fan, C.-L. Xie, Q. Liu, Z.-H. Luo, G. Liu, X.-W. Yang, *J. Nat. Prod.* **2017**, *80*, 2174–2177; g) S. Niu, Z. Fan, X. Tang, Q. Liu, Z. Shao, G. Liu, X.-W. Yang, *Tetrahedron Lett.* **2018**, *59*, 375–378; h) H.-Y. Chen, T.-K. Liu, Q. Shi, X.-L. Yang, *Fitoterapia* **2019**, *137*, 104243; i) Z. Cheng, Y. Li, W. Xu, W. Liu, L. Liu, D. Zhu, Y. Kang, Z. Luo, Q. Li, *Bioorg. Chem.* **2019**, *91*, 103129; j) F. Li, W. Sun, S. Zhang, W. Gao, S. Lin, B. Yang, C. Chai, H. Li, J. Wang, Z. Hu, Y. Zhang, *Chin. Chem. Lett.* **2020**, *31*, 197–201.

[2] a) S.-H. Hou, Y.-Q. Tu, S.-H. Wang, C.-C. Xi, F.-M. Zhang, S.-H. Wang, Y.-T. Li, L. Liu, *Angew. Chem. Int. Ed.* **2016**, *55*, 4456–4460; *Angew. Chem.* **2016**, *128*, 4532–4536; b) P. Hu, H. M. Chi, K. C. DeBacker, X. Gong, J. H. Keim, I. T. Hsu, S. A. Snyder, *Nature* **2019**, *569*, 703–707; c) V. Singh, R. B. Singh, S. M. Mobin, *Tetrahedron* **2009**, *65*, 7969–7974; d) T. K. Behera, D. B. Jarhad, S. M. Mobin, V. Singh, *Tetrahedron* **2016**, *72*, 5377–5393; e) M. J. Kim, S. Lee, T. Kang, M.-H. Baik, H.-Y. Lee, *Eur. J. Org. Chem.* **2020**, 609–617.

[3] T. Shiina, K. Nakagawa, Y. Fujisaki, T. Ozaki, C. Liu, T. Toyomasu, M. Hashimoto, H. Koshino, A. Minami, H. Kawaide, H. Oikawa, *Biosci. Biotechnol. Biochem.* **2019**, *83*, 192–201.

[4] a) G. Mehta, A. Srikrishna, *Chem. Rev.* **1997**, *97*, 671–720; b) T. Hudlicky, J. D. Price, *Chem. Rev.* **1989**, *89*, 1467–1486.

[5] A. W. Schmidt, H. J. Knölker, *Synlett* **2010**, 2207–2239.

[6] a) R. L. Danheiser, D. J. Carini, A. Basak, *J. Am. Chem. Soc.* **1981**, *103*, 1604–1606; b) R. L. Danheiser, D. J. Carini, D. M. Fink, A. Basak, *Tetrahedron* **1983**, *39*, 935–947; c) R. L. Danheiser, D. M. Fink, *Tetrahedron Lett.* **1985**, *26*, 2513–2516; d) R. L. Danheiser, C. A. Kwasigroch, Y. M. Tsai, *J. Am. Chem. Soc.* **1985**, *107*, 7233–7235; e) D. A. Becker, R. L. Danheiser, *J. Am. Chem. Soc.* **1989**, *111*, 389–391; f) R. L. Danheiser, E. J. Stoner, H. Koyama, D. S. Yamashita, C. A. Klade, *J. Am. Chem. Soc.* **1989**, *111*, 4407–4413; g) M. Suginome, S.-i. Matsunaga, Y. Ito, *Synlett* **1995**, 941–942.

[7] a) J. C. Friese, S. Krause, H. J. Schäfer, *Tetrahedron Lett.* **2002**, *43*, 2683–2685; b) G. V. Saborit, C. Cativiela, A. I. Jiménez, J. Bonjoch, B. Bradshaw, *Beilstein J. Org. Chem.* **2018**, *14*, 2597–2601.

[8] a) K. D. Schwartz, J. D. White, *Org. Synth.* **2006**, *83*, 49–54; b) J. G. Millar, S. L. Midland, *Tetrahedron Lett.* **2007**, *48*, 6377–6379.

[9] a) E. J. Corey, C. J. Helal, *Angew. Chem. Int. Ed.* **1998**, *37*, 1986–2012; *Angew. Chem.* **1998**, *110*, 2092–2118; b) E. J. Corey, R. K. Bakshi, S. Shibata, C. P. Chen, V. K. Singh, *J. Am. Chem. Soc.* **1987**, *109*, 7925–7926.

[10] A. M. Martín Castro, *Chem. Rev.* **2004**, *104*, 2939–3002.

[11] a) O. H. Oldenziel, A. M. van Leusen, *Tetrahedron Lett.* **1973**, *14*, 1357–1360; b) U. Schöllkopf, R. Schröder, *Angew. Chem. Int. Ed. Engl.* **1973**, *12*, 407–408; *Angew. Chem.* **1973**, *85*, 402–403; c) O. H. Oldenziel, D. Van Leusen, A. M. Van Leusen, *J. Org. Chem.* **1977**, *42*, 3114–3118; d) A. M. van Leusen, P. G. Oomkes, *Synth. Commun.* **1980**, *10*, 399–403.

 © 2020 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

[12] a) S. W. M. Crossley, C. Obradors, R. M. Martinez, R. A. Shenvi, *Chem. Rev.* **2016**, *116*, 8912 – 9000; b) R. W. Hoffmann, *Chem. Soc. Rev.* **2016**, *45*, 577 – 583; c) S. A. Green, S. W. M. Crossley, J. L. M. Matos, S. Vásquez-Céspedes, S. L. Shevick, R. A. Shenvi, *Acc. Chem. Res.* **2018**, *51*, 2628 – 2640; d) O. J. Turner, J. A. Murphy, D. J. Hirst, E. P. A. Talbot, *Chem. Eur. J.* **2018**, *24*, 18658 – 18662.
[13] S. Stanković, J. H. Espenson, *J. Org. Chem.* **1998**, *63*, 4129 – 4130.
[14] P. A. Grieco, S. Gilman, M. Nishizawa, *J. Org. Chem.* **1976**, *41*, 1485 – 1486.
[15] a) T. F. Jamison, S. Shambayati, W. E. Crowe, S. L. Schreiber, *J. Am. Chem. Soc.* **1994**, *116*, 5505 – 5506; b) T. F. Jamison, S. Shambayati, W. E. Crowe, S. L. Schreiber, *J. Am. Chem. Soc.* **1997**, *119*, 4353 – 4363.
[16] a) K. M. Nicholas, J. Siegel, *J. Am. Chem. Soc.* **1985**, *107*, 4999 – 5001; b) D. F. McComsey, A. B. Reitz, C. A. Maryanoff, B. E. Maryanoff, *Synth. Commun.* **1986**, *16*, 1535 – 1549.
[17] S. Shambayani, W. E. Crowe, S. L. Schreiber, *Tetrahedron Lett.* **1990**, *31*, 5289 – 5292.
[18] M. Huang, *J. Am. Chem. Soc.* **1946**, *68*, 2487 – 2488.
[19] a) L. Caglioti, M. Magi, *Tetrahedron* **1963**, *19*, 1127 – 1131; b) L. Caglioti, *Tetrahedron* **1966**, *22*, 487 – 493; c) R. O. Hutchins, B. Maryanoff, C. Milewski, *J. Am. Chem. Soc.* **1971**, *93*, 1793 – 1794; d) G. W. Kabalka, D. T. C. Yang, J. D. Baker, *J. Org. Chem.* **1976**, *41*, 574 – 575.
[20] D. H. R. Barton, S. W. McCombie, *J. Chem. Soc. Perkin Trans. 1* **1975**, 1574 – 1585.
[21] G. A. Molander, G. Hahn, *J. Org. Chem.* **1986**, *51*, 2596 – 2599.

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